Data Science

Capital Compensation

Noëlle Crawford

AGENDA

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Dataset Overview

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Source: Washington DC Open Data Portal (Public Salary Records).
Scale: Over 11,000+ individual employee records.
Features: Includes Agency Name, Job Title, Grade, Salary, and Start Date.

First 10 Rows

FIRST_NAME	LAST_NAME	JOBTITLE	DESCRSHORT	GRADE	COMPRATE	HIREDATE_STRING	GVT_TYPE_OF_APPT	OBJECTID
Natasha	Boley	Compliance Specialist	DBH	13	123873.00	2004/10/18	CS - Reg	1
Korlu	Wolobah-Kuyon	PSYCHIATRIC NURSE	DBH	9	115907.00	2020/08/17	CS - Reg	2
Hadizat	Elayo-Ross	SUPERVISORY PSYCHIATRIC NURSE	DBH	13	122500.00	2023/08/14	MSS - Reg	3
Gary	Evans	Climate Control Maint Mechanic	DBH	11	41.73	1985/08/27	CS - Reg	4
Jemia	Gross	Special Police Officer	DBH	8	75787.00	2016/11/14	CS - Reg	5
James	Broughton	Program Monitor	DBH	12	107046.00	1987/10/01	CS - Reg	6
Gertrude	Allen	Program Support Assistant	DBH	7	67548.00	1990/12/03	CS - Reg	7
Karen	Bowie	Behavioral Health Technician	DBH	8	63711.00	2021/05/24	CS - Reg	8
Jeremy	Park	MEDICAL OFFICER PSYCH RES	DBH	0	70325.00	2025/07/01	CS - Temp	9
Richard	Gontang	Chief Clinical Officer	DBH	15	177839.89	1994/11/03	MSS - Reg	10

Objective

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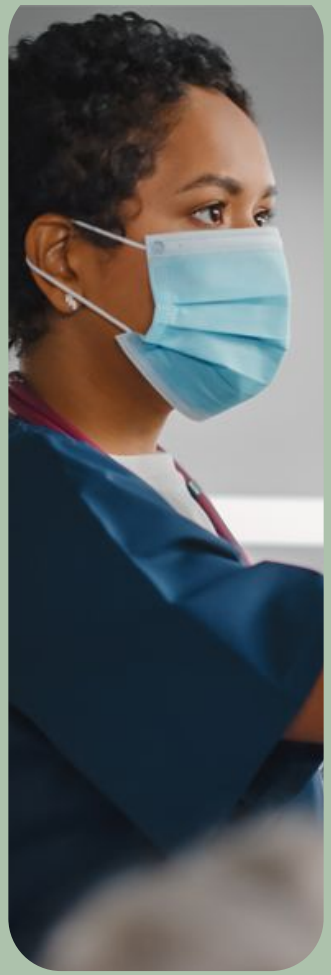
The Mission: To determine how salary distributions correlate with agency size, professional grade levels, and appointment types across DC’s public infrastructure.
Goal: Identify the primary drivers of compensation in the capital's public workforce.



Standardization

- Addressed non-standard job titles to group similar roles.

Filtering

- Focused on major agencies to eliminate "noise" from micro-agencies.

Tools Used

- Python (Pandas for cleaning, Matplotlib/Seaborn for visualization).

3 Data Cleansing and Preparation

Key Analysis Questions

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Distribution: Which agencies command the highest average salaries versus the largest total workforce?
Correlation: Does a higher "Grade" level consistently translate to a linear increase in salary across different agencies?
Structure: How does "Appointment Type" (e.g., Educational vs. Career) impact the pay floor and ceiling?

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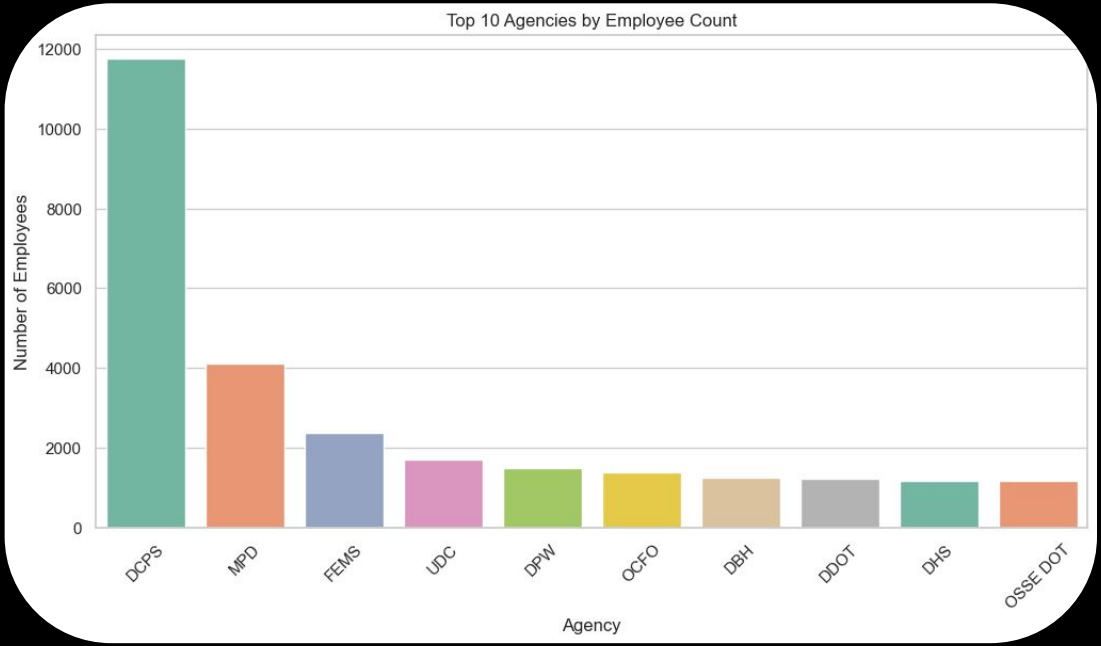
Visualization - Work Force Scale

Insight: DCPS dominates the dataset with over 11,000 employees, dwarfing other agencies in terms of headcount.



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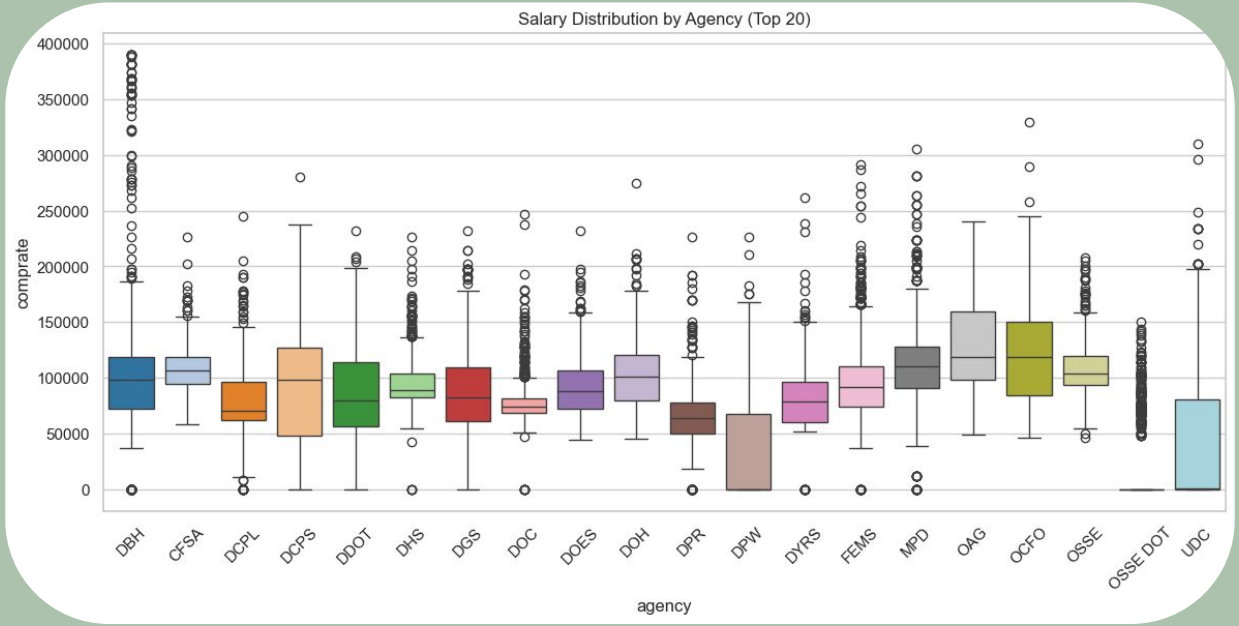
Visualization - Salary Distribution

Insight: Specialized agencies like the Office of the Chief Financial Officer (OCFO) and the Office of the Attorney General (OAG) show higher average salary clusters compared to the broader DCPS workforce.



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Visualization - Grade vs. Compensation

Insight: We do not see as strong of a correlation as expected. But that's because the different appointment types get paid differently based off grade.

DCPS Scale

- **DCPS Scale:** DCPS is the largest employer, but its salary distribution is wide and varied.

High-Value Sectors

- **High-Value Sectors:** OCFO, OAG, and MPD tend to have higher average individual salaries.

Primary Drivers

- **Primary Drivers:** Salary is most heavily influenced by an employee's Grade level and their specific Appointment Type.

8 Key Findings

Limitations	Lack of demographic data (race, gender) prevents a complete equity analysis; some job titles remain non-standardized.
Future Directions	Implementing predictive modeling to estimate salaries and using clustering to group similar job roles across diverse agencies.
Future Directions	Compare these salaries to what people in the private sector earn for the same roles. This would help determine if the DC government is competitive enough to attract and retain top-tier talent in a high-cost region.

Limitations and Future Decisions

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Conclusion

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Final Transmission: Public salary in DC is a balance between the massive educational workforce (DCPS) and high-paying specialized legal and financial sectors.
Takeaway: Grade and Appointment type are the most reliable predictors of salary in the current dataset.

Any Questions?

THANK YOU

Noëlle Crawford